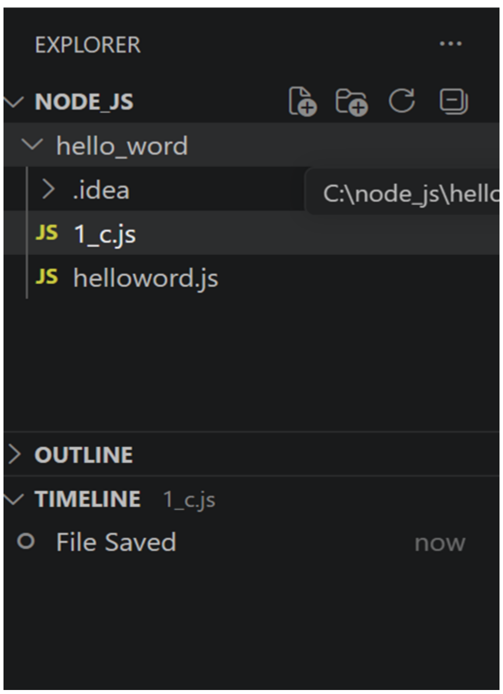
**B.Sc. (Information Technology) (Sem.- V)**

# Name of the Course: Full Stack development MERN Practical

**Practical 1. Environment Setup and JavaScript Basics**

a. Install Node.js and Visual Studio Code, and verify installation.
b. Create a simple Node.js file and run it using the terminal.
c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).
d. Create and run a simple program using variables and functions.
e. Demonstrate use of operators and control statements in JavaScript
f. Write programs using loops and template literals.

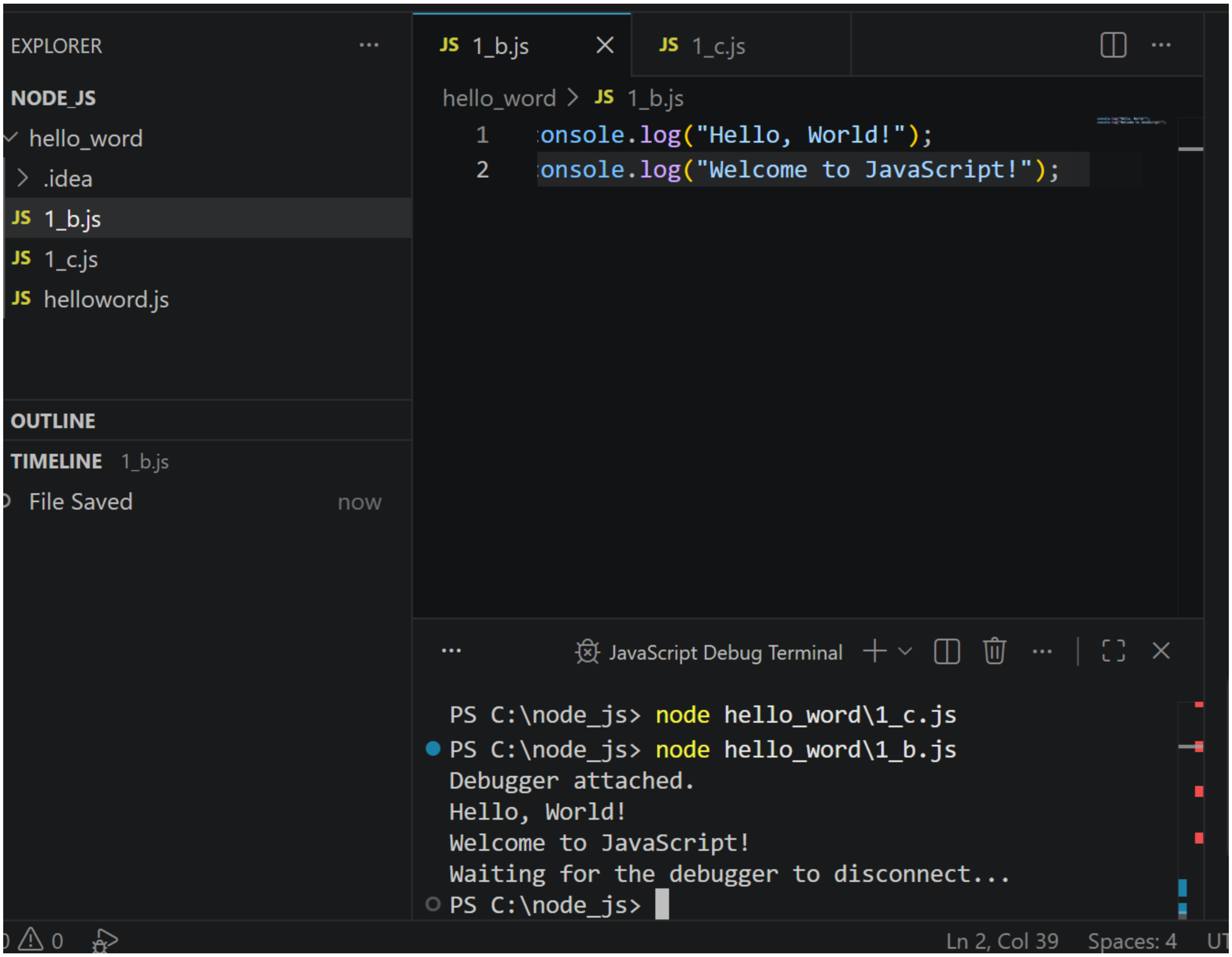
**b) Create a simple Node.js file and run it using the terminal.**
**Solution** :
Step 1: Create folder with name node_js
Step 2: Open folder in vs code
Step 3:
On explorer -> select node_js folder -> create new file -> Give file name with extension .js




Write Program on vs code

```
console.log("Hello, World!");
console.log("Welcome to JavaScript!");
```

Step 4: On terminal run it

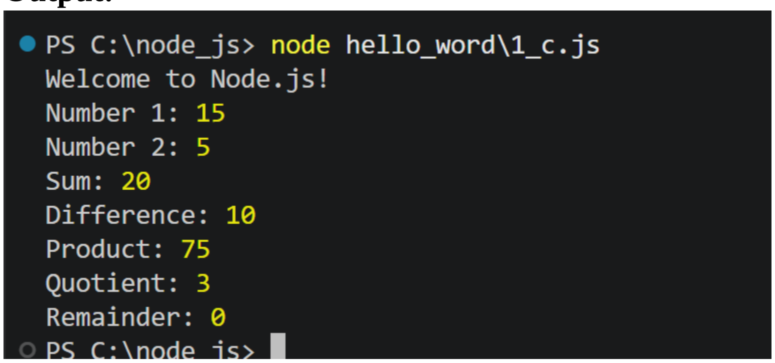



**c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).**

```
// Basic Node.js Script

// Display a message
console.log("Welcome to Node.js!");

// Declare variables
let num1 = 15;
let num2 = 5;

// Perform arithmetic operations
let sum = num1 + num2;
let difference = num1 - num2;
let product = num1 * num2;
```

```
let quotient = num1 / num2;
let remainder = num1 % num2;

// Display variable values
console.log("Number 1:", num1);
console.log("Number 2:", num2);

// Display arithmetic results
console.log("Sum:", sum);
console.log("Difference:", difference);
console.log("Product:", product);
console.log("Quotient:", quotient);
console.log("Remainder:", remainder);
```

**Output:**




**d. Create and run a simple program using variables and functions.**

**Solution:**

```
let num1 = 15;
let num2 = 25;

// Function to add two numbers
function add(a, b) {
  return a + b;
}

// Store the result
let result = add(num1, num2);

// Display the result
console.log("First Number:", num1);
console.log("Second Number:", num2);
console.log("Sum:", result);
```

**Output:**

```
PS C:\node_js> node hello_word\1_d.js;
PS C:\node_js> node hello_word\1_d.js;
Debugger attached.
First Number: 15
Second Number: 25
Sum: 40
Waiting for the debugger to disconnect...
PS C:\node_js>
```

```
// Import readline module to take input from the user
const readline = require("readline");

// Create readline interface
const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout
});

// Function to display user details
function displayDetails(name, age) {
    console.log("\nUser Details");
    console.log("Name: " + name);
    console.log("Age: " + age);
}

// Get user input
rl.question("Enter your name: ", function(name) {

    rl.question("Enter your age: ", function(age) {

        // Call the function
        displayDetails(name, age);

        // Close readline
        rl.close();
    });

});
```

```
PS C:\node_js> node hello_word\1_d.js;
Debugger attached.
Enter your name: Minakshi
Enter your age: 23

User Details
Name: Minakshi
Age: 23
Waiting for the debugger to disconnect...
PS C:\node_js>
```

**e. Demonstrate use of operators and control statements in JavaScript**

**Solution:**

**Operators and Control Statements:**

```
// Declare variables
let num1 = 20;
let num2 = 10;

// Arithmetic Operators
console.log("Addition = " + (num1 + num2));
console.log("Subtraction = " + (num1 - num2));
console.log("Multiplication = " + (num1 * num2));
console.log("Division = " + (num1 / num2));

// Relational Operator and Control Statement
if (num1 > num2) {
   console.log(num1 + " is greater than " + num2);
} else if (num1 < num2) {
   console.log(num1 + " is less than " + num2);
} else {
   console.log("Both numbers are equal");
}
```

**Output:**

```
PS C:\node_js> node hello_word\1_e.js;
Debugger attached.
Addition = 30
Subtraction = 10
Multiplication = 200
Division = 2
20 is greater than 10
Waiting for the debugger to disconnect...
```

**Example with User input:**

```
const readline = require("readline");
```

```
const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

rl.question("Enter first number: ", function(num1) {

  rl.question("Enter second number: ", function(num2) {

    num1 = Number(num1);
    num2 = Number(num2);

    console.log("Addition = " + (num1 + num2));
    console.log("Subtraction = " + (num1 - num2));
    console.log("Multiplication = " + (num1 * num2));
    console.log("Division = " + (num1 / num2));

    if (num1 > num2) {
      console.log(num1 + " is greater than " + num2);
    } else if (num1 < num2) {
      console.log(num1 + " is less than " + num2);
    } else {
      console.log("Both numbers are equal");
    }

    rl.close();
  });

});
```

**Output:**

```
Debugger attached.
Enter first number: 10
Enter second number: 20
Addition = 30
Subtraction = -10
Multiplication = 200
Division = 0.5
10 is less than 20
Waiting for the debugger to disconnect...
PS C:\node_js>
```

**f. Write programs using loops and template literals.**

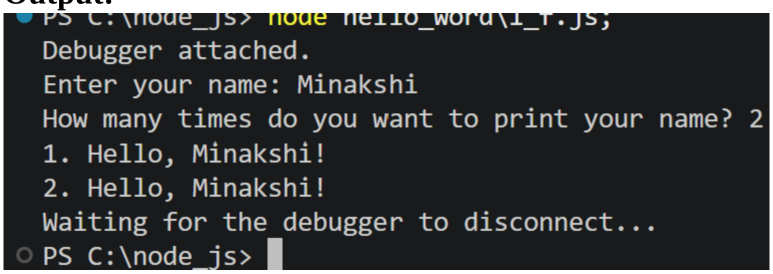
**Solution:**

**For Loop**

```
const readline = require("readline");
```

```
const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout
});

rl.question("Enter your name: ", function(name) {

    rl.question("How many times do you want to print your name? ", function(n) {

        for (let i = 1; i <= Number(n); i++) {
            console.log(`${i}. Hello, ${name}!`);
        }

        rl.close();
    });

});
```

**Output:**




**While loop**

```
let i = 1;
while (i <= 10)
 {
console.log(‘2 x ${i} = ${2 * i}’);
i++;
}
```

**Output:**

```
PS C:\node_js> node hello_word\1_f.js;
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
Waiting for the debugger to disconnect...
```
,

**<u>Practical 2. Core JavaScript Concepts for MERN</u>**

a. Create functions and call them with parameters.
b. Work with arrays and objects in JavaScript.
c. Demonstrate use of arrow functions.
d. Write a program using array methods (map, filter, reduce).
e. Create a program that manipulates objects and displays output.
f. Implement a small JavaScript program combining functions, arrays, and objects.

**a. Create functions and call them with parameters.**

```
// Import readline module
const readline = require("readline");

const rl = readline.createInterface({
    input: process.stdin,
    output: process.stdout
});

// Function 1: Display Heading
function displayHeading() {
    console.log("========================================");
    console.log("     STUDENT RESULT MANAGEMENT");
    console.log("========================================");
}

// Function 2: Calculate Total Marks
function calculateTotal(m1, m2, m3) {
    return m1 + m2 + m3;
}

// Function 3: Calculate Percentage
function calculatePercentage(total) {
    return total / 3;
```

```
}

// Function 4: Calculate Grade
function calculateGrade(percentage) {
    if (percentage >= 90)
        return "A+";
    else if (percentage >= 75)
        return "A";
    else if (percentage >= 60)
        return "B";
    else if (percentage >= 50)
        return "C";
    else if (percentage >= 35)
        return "D";
    else
        return "Fail";
}

// Function 5: Display Student Result
function displayResult(name, rollNo, total, percentage, grade) {
    console.log("\n========== RESULT ==========");
    console.log("Student Name : " + name);
    console.log("Roll Number  : " + rollNo);
    console.log("Total Marks  : " + total + "/300");
    console.log("Percentage   : " + percentage.toFixed(2) + "%");
    console.log("Grade        : " + grade);
    console.log("============================");
}

// Main Program
displayHeading();

rl.question("Enter Student Name: ", function(name) {

    rl.question("Enter Roll Number: ", function(rollNo) {

        rl.question("Enter Marks in Subject 1: ", function(sub1) {

            rl.question("Enter Marks in Subject 2: ", function(sub2) {

                rl.question("Enter Marks in Subject 3: ", function(sub3) {

                    // Convert string input to number
```

```
                let marks1 = Number(sub1);
                let marks2 = Number(sub2);
                let marks3 = Number(sub3);

                // Call functions
                let total = calculateTotal(marks1, marks2, marks3);
                let percentage = calculatePercentage(total);
                let grade = calculateGrade(percentage);

                displayResult(name, rollNo, total, percentage, grade);

                rl.close();
            });
        });
    });
  });
});
```

**Output:**

```
PS C:\node_js> node hello_word\2_a.js;
Enter Marks in Subject 2: 12
Enter Marks in Subject 3: 12

========== RESULT ==========
Student Name : Minakshi
Roll Number  : 1
Total Marks  : 36/300
Percentage   : 12.00%
Grade        : Fail
============================
```

**b. Work with arrays and objects in JavaScript.**

**Solution:**

```
const readline = require("readline");

const rl = readline.createInterface({
  input: process.stdin,
  output: process.stdout
});

// Array to store student objects
let students = [];

// Function to add student
function addStudent(name, roll, marks) {
  let student = {
```

```
        name: name,
        roll: roll,
        marks: marks
    };

    students.push(student);
}

// Function to display all students
function displayStudents() {
    console.log("\n========== STUDENT DETAILS ==========");

    for (let i = 0; i < students.length; i++) {
        console.log("Student " + (i + 1));
        console.log("Name  : " + students[i].name);
        console.log("Roll  : " + students[i].roll);
        console.log("Marks : " + students[i].marks);
        console.log("-----------------------------");
    }
}

// Accept input for Student 1
rl.question("Enter Student 1 Name: ", function(name1) {

    rl.question("Enter Student 1 Roll Number: ", function(roll1) {

        rl.question("Enter Student 1 Marks: ", function(marks1) {

            addStudent(name1, roll1, Number(marks1));

            // Accept input for Student 2
            rl.question("Enter Student 2 Name: ", function(name2) {

                rl.question("Enter Student 2 Roll Number: ", function(roll2) {

                    rl.question("Enter Student 2 Marks: ", function(marks2) {

                        addStudent(name2, roll2, Number(marks2));

                        // Display all students
                        displayStudents();

                        rl.close();
```

```
                });
            });
        });
    });
});
});
```

**Output:**

```
PS C:\node_js> node hello_word\2_b.js;
Student 1
Name  : minakshi
Roll  : 1
Marks : 98
-----------------------------
Student 2
Name  : Sneha
Roll  : 90
Marks : 98
```

**c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).**

**Solution:**

```
// ==========================================
// Program: Employee Salary Calculator
// Demonstrates console.log(), variables,
// and arithmetic operations
// ==========================================

// Display Heading
console.log("=====================================");
console.log("      EMPLOYEE SALARY CALCULATOR");
console.log("=====================================");

// Declare variables
let employeeName = "Rahul Sharma";
let employeeID = 1001;
let basicSalary = 30000;
let hra = 5000;
let da = 3000;
let tax = 2500;

// Display employee details
console.log("\nEmployee Details");
console.log("-------------------------");
```

```
console.log("Employee Name :", employeeName);
console.log("Employee ID   :", employeeID);
console.log("Basic Salary  :", basicSalary);
console.log("HRA           :", hra);
console.log("DA            :", da);
console.log("Tax           :", tax);

// Arithmetic operations
let grossSalary = basicSalary + hra + da;
let netSalary = grossSalary - tax;
let monthlyBonus = basicSalary * 0.10;
let yearlySalary = netSalary * 12;
let averageMonthlyIncome = yearlySalary / 12;

// Display results
console.log("\nSalary Details");
console.log("-------------------------");
console.log("Gross Salary         = " + grossSalary);
console.log("Net Salary           = " + netSalary);
console.log("Monthly Bonus (10%)   = " + monthlyBonus);
console.log("Yearly Salary        = " + yearlySalary);
console.log("Average Monthly Income= " + averageMonthlyIncome);

console.log("\nProgram Completed Successfully!");
```

**Output:**

```
PS C:\node_js> node hello_word\2_c.js;
Tax            : 2500

Salary Details
-------------------------
Gross Salary          = 38000
Net Salary            = 35500
Monthly Bonus (10%)   = 3000
Yearly Salary         = 426000
Average Monthly Income= 35500

Program Completed Successfully!
PS C:\node_js>
```

**d. Write a program using array methods (map, filter, reduce).**
**Solution:**

Op: let marks = [45, 78, 90, 56, 32, 88, 67, 95];

**1. map()**

Creates a **new array** by applying a function to each element.

**e.g:** let updatedMarks = marks.map(function(mark) {
return mark + 5;
});
**O/P: [50, 83, 95, 61, 37, 93, 72, 100]**

**2. filter()**
Creates a **new array** containing only the elements that satisfy a condition.
**e.g:** let passedStudents = marks.filter(function(mark) {
return mark >= 50;
});
**O/P: [78, 90, 56, 88, 67, 95]**

**3. reduce():**
Reduces the array to a single value, such as a sum.
**e.g:** let totalMarks = marks.reduce(function(total, mark) {
return total + mark;
}, 0);
**O/P: 551**

```
// ==========================================
// Program: Array Methods - map(), filter(), reduce()
// ==========================================

// Create an array of marks
let marks = [45, 78, 90, 56, 32, 88, 67, 95];

console.log("Original Marks:");
console.log(marks);

// ------------------ map() ------------------
// Add 5 bonus marks to each student

let updatedMarks = marks.map(function(mark) {
   return mark + 5;
});

console.log("\nMarks after adding 5 bonus marks:");
console.log(updatedMarks);

// ---------------- filter() -----------------
// Display only students who passed (marks >= 50)

let passedStudents = marks.filter(function(mark) {
```

```
    return mark >= 50;
});

console.log("\nPassed Students (Marks >= 50):");
console.log(passedStudents);

// ---------------- reduce() -----------------
// Calculate total marks

let totalMarks = marks.reduce(function(total, mark) {
    return total + mark;
}, 0);

console.log("\nTotal Marks = " + totalMarks);

// Calculate average marks
let average = totalMarks / marks.length;

console.log("Average Marks = " + average.toFixed(2));
```

**Output:**

```
PS C:\node_js> node hello_word\2_d.js;
  45, 78, 90, 56,
  32, 88, 67, 95
]

Marks after adding 5 bonus marks:
[
  50, 83, 95,  61,
  37, 93, 72, 100
]

Passed Students (Marks >= 50):
[ 78, 90, 56, 88, 67, 95 ]

Total Marks = 551
Average Marks = 68.88
PS C:\node_js> 
```

**e. Create a program that manipulates objects and displays output.**
**Solution :**

1. **Create an Object:**
   Creates a student object with four properties.
   **e.g:** let student = {
       name: "Sanil",

```
   rollNo: 101,
   course: "B.Sc. Computer Science",
   marks: 85
};
```

2. **Access Object Properties**
   Accesses object values using **dot notation**.
   **e.g:** console.log(student.name);
   console.log(student.rollNo);

3. **Update a Property**
   Changes the value of the marks property.
   **e.g:** student.marks = 92;

4. **Add a New Property**
   Adds a new property city to the object.
   **e.g:** student.city = "Mumbai";

5. **Delete a Property**
   Removes the course property from the object.
   **e.g:** delete student.course;

```
// ==========================================
// Program: Object Manipulation in Node.js
// ==========================================

// Create an object
let student = {
   name: "Minakshi",
   rollNo: 101,
   course: "  B.Sc. Information Technology",
   marks: 85
};

// Display original object
console.log("===== Original Student Details =====");
console.log(student);

// Access object properties
console.log("\nStudent Name :", student.name);
console.log("Roll Number  :", student.rollNo);
```

```
// Update an existing property
student.marks = 92;

// Add a new property
student.city = "Mumbai";

// Delete a property
delete student.course;

// Display updated object
console.log("\n===== Updated Student Details =====");
console.log(student);

// Display each property separately
console.log("\nStudent Name :", student.name);
console.log("Roll Number  :", student.rollNo);
console.log("Marks        :", student.marks);
console.log("City         :", student.city);
```

**Output:**

```
PS C:\node_js> node hello_word\2_e.js;

Student Name : Minakshi
Roll Number  : 101
Marks        : 92
City         : Mumbai
```